

LABORATORY SCOPE

Personnel Qualifications

Name	Title	Qualifications and Experience	Areas Covered

Inspection and Tests

Inspection and Tests	Equipment	Procedure / Method / Standard	Standard	Competency Required Minimum	Record to be Generated
Microhardness (HV)	Leco LM 100AT Microhardness System	Customer Specific Requirements	ASTM E-386	Internal training	Microhardness Report
Rockwell Hardness (B, C)	Tinius Olsen/Newage Rockwell Tester	Customer Specific Requirements	ASTM E-18	Internal training	IPI Form
Metallographic Sample Prep	Leco PR-36 Mount Press/Leco GPX 300 Grinder/Polisher	MA-WI-76-003 Microstructure Preparation	ASTM E3	Internal training	N/A
Metallographic Evaluation	Olympus GX51 Metallography	Customer Specific Requirements	ASTM E883	Internal training	Quality Drive Customer File Photo Documentation
Weld Analysis	Leco MSX 255 Saw/Leco PR-36 Mount Press/Leco GPX 300 Grinder/Polisher/Olympus SZX7/Pax-It Software	NAMEs User Manual/MA-WI-76-003 Microstructure Preparation/MA-WI-76-004 mixture of Etching Solution/	NAMEs User Manual/	Internal training	
Tensile Test	Zwick 100kN Tensile Tester	Customer Specific Requirements	ASTM E8	Internal training	
Bend Test	Instron 3382 100kN Tensile/Compression Tester	Customer Specific Requirements	ASTM A370	Internal training	
Nital Acid Prep	Nitric Acid, Denatured Alcohol, Grad.Cyl, Beaker, Pipette, Storage Bottle	MA-WI-76-004	ASM Metallography & Microstructure Handbook	Internal training	N/A
Depth of Decarburization	Leco PR-36 Mount Press/Leco GPX 300 Grinder/Polisher/Olympus SZX7/Olympus GX51 Metallograph/Pax-It Software	Customer Specific Requirements	ASTM E1077	Internal training	
Pickle Test	Leco MSX Saw/Etch Bath	Customer Specific Requirements	MA-WI-76-006 Pickle Testing	Internal training	
Surface Roughness	Mitutoyo SJ-201P SR Tester (R _a /R _z)	Customer Specific Requirements	SAE J911		

References

Reference Name	Reference Number
Control of Records	WI 5001
WI - Microstructure Preparation	
Mixture of Etching Solution	
Tensile Testing	
Pickle Test	
Rockwell Test	
Weld Analysis	
Bend Test	
Depth of Decarburization	
Surface Roughness	
FCA MS.50001	
Ford WSS-M99P39-A1	
GMW16579	

Records

Record Name	Record Number	Retention Period
Microhardness Test		7 years?
Tensile Test		

Revision History

Revision Level	Revision Date	Change
A		Release of new document